

# SHRINIVASAN S G

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## Education

**B.Tech. Artificial Intelligence and Data Science**  
Chennai Institute of Technology

09/2023 – Present  
CGPA: 9.17

## Professional Experience

- Prodigy Infotech Pvt.Ltd.,** Software Development Intern 11/2024-12/2024
- Developed scalable Python applications using algorithmic optimization, clean architecture, and efficient code deployment practices.
  - Utilized core Python libraries for **API integration**, data handling, UI responsiveness, and performance enhancement.
  - Implemented modular design, Git version control, and **SDLC** best practices for maintainable, scalable software solutions.
- Chennai Institute of Technology,** Research Internship 11/2024-12/2024
- Applied **ML models** for air quality prediction, pollution analysis, and environmental data insights generation improvement.
  - Designed **hybrid ML and DL** architectures for accurate flood forecasting and disaster warning system reliability.
  - Improved model accuracy through feature engineering, preprocessing, hyperparameter tuning, and performance evaluation metrics.

## Projects

- RAIL MADAD COMPLAINT SYSTEM,** MERN stack | Huggingface | Python | Postman
- Built a **Full-Stack** system using **Node.js** and **MongoDB** for **structured grievance submission, tracking, and incident resolution.**
  - Integrated **NLP models** for **automated classification, keyword extraction, and chatbot support** to **streamline** passenger actions.
  - Engineered **RESTful APIs** and interactive data dashboards to enhance scalability and operational decision-making.
- AI DRIVEN DISASTER MANAGEMENT,** MERN Stack | Tensorflow | Pytorch
- Deployed an **AI-driven platform** using **Random Forest models** to predict **cyclones, floods, and earthquakes** with **90%+ accuracy.**
  - Integrated **real-time monitoring** and **NLP chatbots** into a **Full-Stack architecture** to enhance **citizen emergency response.**
  - Enhanced **decision-making** by **automating alerts and risks** to enable **rapid support** for **authorities and communities.**
- Unauthorized Billboard Detector,** MERN stack | Opencv | YOLO | Tensorflow | Pytorch
- Utilized **OpenCV** for **image preprocessing** and **contour detection** to isolate **billboard regions** from **complex surveillance feeds.**
  - Trained **YOLO models** using **TensorFlow** and **PyTorch** to achieve **accurate real-time identification** of **unauthorized billboards.**
  - Annotated datasets with **LabelImg** and optimized **deployment workflows** to improve **precision** for **large-scale monitoring.**

## Certifications

**AWS Cloud Practitioner(SAA CO2):** Cloud Computing: AWS | **CCNA: Introduction to Networks:** Cisco Networking Academy | **Cybersecurity Essentials:** Cisco Networking Academy | **Networking Essentials:** Cisco Networking Academy | **Database Management System:** NPTEL: IIT-Kharagpur | **Cloud Computing:** NPTEL: IIT-Kharagpur | **Parallel Computer Architecture:** NPTEL: IIT-Kharagpur | **Reinforcement Learning:** NPTEL: IIT-Kharagpur | **Python Essentials:** Cisco Networking Academy | **MongoDB Basics:** Credly

## Skills

**Frontend Development:** HTML5, CSS3, JavaScript (ES6+), React.js, Responsive Design, Tailwind CSS, Bootstrap | **Backend Development:** Node.js, Express.js, Django, Flask, Spring Boot, RESTful APIs, Authentication, API Security | **Databases:** MySQL, PostgreSQL, MongoDB, Redis, Cassandra | **DevOps & Cloud:** AWS (EC2, S3, IAM), Docker, Kubernetes, Git, GitHub, CI/CD Pipelines, Terraform | **System Design & Engineering:** Data Structures & Algorithms, OOP, Design Patterns, Scalability, Caching, Load Balancing | **Testing & Debugging:** Unit Testing (Jest, PyTest, JUnit), API Testing (Postman, Swagger), Logging, Error Handling | **Programming:** JavaScript, Python, Java, SQL, HTML, CSS, C++, C | **Machine Learning:** RNN, CNN, Transformers, Clustering, Model Optimization, DL.

## Achievements

- Selected for the **ideation round** of **SIH 2024** demonstrating **innovative problem-solving** and **technical feasibility skills.**
- Actively engaged in the **AICTE APF program** and **IIC Regional meet** to foster **entrepreneurship** and **innovation.**
- Secured qualification for the **Technova hackathon** qualifier round by demonstrating **rapid prototyping** and **coding skills.**
- Advanced to **Round 2** of **TCS CodeVita** securing a **Global Rank of 3000** demonstrating exceptional **algorithmic proficiency.**

## Publications

- Published a **review paper** titled "**Developing Sustainable Environment Against Air Pollution Impact Using AI**" in **IEEE Xplore.**

## Coding Profile

- Achieved a **2-Star rating** on **CodeChef** by solving **300+ algorithmic problems** using **optimized data structures.**
- Solved **850+ LeetCode problems** achieving a **max rating of 1700+** through **consistent algorithmic practice** and **optimization.**
- Earned **136 bronze badges** on **Skillrack** by solving **450+ coding challenges** across various **domains** and **difficulty levels.**
- Coding:** Leetcode: Rating-1779, Problems solved:850 | Codechef: Rating-1411, Problems solved:300 | Skillrack: Problems solved:485

## Volunteering

- Orchestrated the **logical loop event** during the **department symposium** managing **strategic planning, team coordination.**
- Contributed to **Sustainable Development Goals** by volunteering for a **tree-planting initiative** organized in **Puduper village.**